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Hydrogen production using plasma gasification with steam injection

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ABSTRACT

Plasma gasification is a promising gasification technology intended at providing sustainable disposal for various wastes. In this work, a process model was developed to simulate the biomass plasma gasification using Aspen Plus simulator. Effects of critical parameters, including gasification temperature, Equivalence Ratio (ER) and Steam-to-Biomass Ratio (SBR) on the composition of fuel gas were discussed. The model is validated against experimental data and found to be in good agreement. The results indicate that low temperatures are more favourable for the production of hydrogen, while high ER has a negative effect on the hydrogen production. The simulation results also demonstrate that steam injection is a key factor to produce more hydrogen rich gas in the SBR range studied, but had a major effect on CO_2 formation. The temperature and the SBR show opposite behavior on the syngas LHV, which is attributed to the CO content in the syngas that increases with temperature and decreases with SBR. Results of plasma gasification show similar syngas LHV trends for the three biomasses cases being the higher syngas LHV obtained for vines pruning. These data are crucial to describe scenarios concerning the potential use of biomass as energy source.

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Introduction

With the last 2015 Paris Climate Change Conference setting new goals for CO_2 and pollution gases new alternative energies are in the up rise. Fossil fuels besides being more depleted every day [1] contribute directly to the formation of greenhouse gases [2] so they need to be change for a more renewable and cleaner fuel. To address this problem gasification has become a major interest in recent years to a great number of researches attracted to the advantages of using different kinds of feedstock [3]. Gasification is a thermochemical process that converts raw fuel into a fuel gas [4]. This fuel gas is mainly composed by carbon monoxide (CO), hydrogen (H_2), water vapor (H_2O), carbon dioxide (CO_2), tar and ash. After filtration to remove tars and ashes, the final product is obtained, syngas, which can be used as fuel. This synthesis gas, as is also referred, is converted from a variety of fuels being the most common: coal, biomass, plastic and municipal solid wastes [5]. Biomass gasification has been transformed into one of the most promising methods in the production of syngas [6]. Biomass is and abundant resource, low in nitrogen

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and sulphur content [7], due to its formation nature is also seen as a carbon neutral fuel, not adding CO_2 emissions to the environment [8].

The gasification is characterized by four phases, namely: drying, withdrawing water present in the fuel; pyrolysis, characterized by the decomposition of volatile matters; combustion, needed to supply heat to the system; and finally the gasification reactions which occur originating the final product, syngas.

An important evolution in gasification technology came with the introduction of plasma torches during the process [9,10]. The plasma provides high temperature heat to the system, while completely decompose the organic portion of the fuel into constituent elements and partly inorganic vitrified slag.

Plasma is the fourth state of matter, warming a solid forms a liquid, heating a fluid forms a gas, heating a gas creates a plasma [5]. During the heating of the gas particles collide with each other, ionizing, and forming free electrons and ions, giving the plasma properties such as the ability to conduct electric current [6]. About 99% of all the matter in the known universe is in the plasma state [5].

The heart of the gasification process with plasma lies in the plasma torch where an intense electric arc is created between two electrodes, which are crossed by a gas [7]. Despite the arc temperature being extremely high (roughly 13,000 °C), the temperature where contact occurs with the fuel is much lower, between 2700 °C and 4500 °C [7]. This temperature is sufficient to effect pyrolysis of complex hydrocarbons in simple gases such as CO and H_2, whilst all inorganic components (glass, metal, silicones, heavy metals) are melted in a volcanic lava witch after cooling vitrifies, the gas produced leaves the gasifier at about 1000 °C–1200 °C [7]. A gasifier scheme with plasma capability can be observed in Fig. 1.

Due to the high temperatures the used feedstock react in the gasifier decomposing into its most basic elements, even hazardous waste can be transformed into useful syngas [8]. The inorganic part that ultimately vitrifies has the advantage of not being dangerous and can be used in various applications such as building material for roads and roofs [8].

The merits and demerits of the plasma gasification were been well expressed by Janajreh et al. [11] in a comparative study

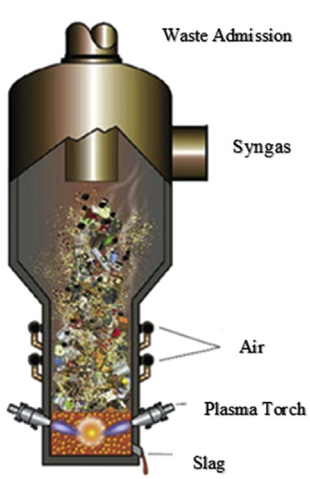

Fig. 1 – Plasma gasifier.

of the conventional gasification and plasma gasification. The high temperature of the plasma process destroys all chemical bonds leading to large amounts of radicals, ions, electrons, and excited molecules in the system. These active species, and the high radiation intensity from the plasma, has the merit to substantially increase the rates of reactions, allow reactions to proceed that are otherwise difficult to obtain under conventional lower temperature conditions [12]. An important demerit of thermal plasma is the high electric energy requirement, which penalizes the process efficiency. The average efficiency was calculated to be about 42% for the plasma gasification and about 72% for the conventional gasification [11]. However, a complete comparative cost evaluation generally demonstrates the economic viability of the plasma technology [13].

Plasma gasification has already attracted interest from many researchers around the world. Some research carried out by Messerle et al. [14] which compared the coal gasification with plasma through experimental and computational simulations. They concluded that both simulations have consistent values, validating the study of plasma gasification by computational means.

Another research prepared by Ojha et al. [15], studied the advantages of using central plasma gasification for treating municipal solid waste in developing countries. As described, advantages are: the elimination of toxic products safely and effectively producing syngas as a greener approach to energy production from an environmental point of view.

Hlina et al. [16] performed the plasma gasification of wood, waste plastics and pyrolysis oil in the reactor equipped by H_2O/Ar plasma torch with extreme properties of plasma. The plasma properties, mainly the high temperature, low mass flow rate, and composition (hydrogen–oxygen plasma with a small amount of Ar) lead to the production of high quality syngas with the high content of CO and H_2 and insignificant content of tars. The major drawback is the higher electricity demand.

Cho et al. [17] used an hybrid gasification system which is divided into a gasification reactor and a plasma reactor to conduct the gasification of a high density polyethylene. They obtained an high yield of syngas in virtue of enhanced gasification reactions in the thermal plasma reactor, while the gasification reactor was operated at the durable condition of atmospheric pressure and low temperature of about 900 K. The hybrid gasification system also allows achieving similar cold gas efficiencies to the conventional gasification system.

The aim of this work is to create a computer model capable of simulating the gasification process with plasma. The software used was Aspen Plus since it does not require massive computational power. The developed model, after validation, is applied in a study of three gasification parameters, the equivalent ratio (ER), the steam to biomass ratio (SBR) and gasifier temperature. Three types of biomass where used: coffee husks, vines pruning and forest residues. The lower heating value (LHV) of the gasification gas product and the H_2 production for each parameter is also analyzed.

Technical approach and process simulation

The model for the biomass plasma gasification process was developed using the software Aspen Plus. This software can

be defined as a process simulation program which provides estimates for parameters of a process such as temperature, composition and sizing under specific conditions by using mass and energy balances relations and phase equilibrium [18]. It consists of many built-in unit operations such as reactors, heat exchangers, separators, distillation and physical properties of the materials. Moreover, this simulation can handle unconventional materials such as solids which are used in many processes including gasification. Adopting proper thermodynamic data, realistic operating conditions and specific equipment models, real process performance can be approached. The great advantage of Aspen Plus over other software able to simulate the plasma gasification process comes from its lower computational capacity requirements.

For the simulation of the gasification process it was divided in four parts, namely: fuel inlet, drying, pyrolysis and gasification. A schematic summary of the steps leading to the creation of the model can be seen in Fig. 2.

The plasma gasification model uses Gibbs free energy minimization as a method to calculate syngas composition. The numerical method is described for the gasification process by Jarungthammachote and Dutta [19]:

At the state of equilibrium, the Gibbs free energy of the system is minimized and defined by:

$$G^t = \sum_{i=1}^{N} n_i \mu_i \tag{1}$$

where n_i and μ_i are the number of moles and chemical potential of species i, respectively. Considering ideal gas behavior for the gases:

$$\mu_i = G_i^0 + RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{2}$$

where R and T are the universal gas constant and temperature. f_i denotes the fugacity of species i. The superscript 0 stands for standard thermodynamic quantity, thus G_i^0 and f_i^0 are the standard Gibbs free energy and the standard fugacity of species i. Eq. (2) expressed in terms of pressure as follows:

$$\mu_i = G_i^0 + RT \ln\left(\frac{\varnothing P_i}{P^0}\right) \tag{3}$$

where ∅ is the fugacity coefficient. Usually f and P can have the same value as the pressure approaches zero. In these conditions, the real gas approaches the ideal gas. If all gases are assumed as ideal at 1 atm, then Eq. (3) can be rewritten as:

$$\mu_i = \Delta G_{f,\,i}^0 + RT \ln(y_i) \tag{4}$$

where y_i represents the molar fraction of gas species i. $\Delta G_{f,\,i}^0$ represents the standard Gibbs free energy of formation of species i, which is set to zero for all chemical elements. Introducing Eq. (4) in Eq. (1) the following is obtained:

$$G^t = \sum_{i=1}^{N} n_i \Delta G_{f,i}^0 + \sum_{i=1}^{N} n_i RT \ln\left(\frac{n_i}{n_{tot}}\right) \tag{5}$$

The objective of the problem is to find the value of set of n_i for which total Gibbs free energy of the system is minimum, subjected to the elemental balance constraint given by:

$$\sum_{i=1}^{N} a_{ij} n_i = A_j \tag{6}$$

where a_{ij} is the number of atoms of the j_{th} element in a mole of i_{th} species and A_j is the total number of atoms of the j_{th} element in the mixture. By applying Lagrange multiplier method the Gibbs free energy of the system can be minimized using the following expression [20]:

$$\frac{\partial L}{\partial n_i} = \Delta G_{f,i}^0 + n_i RT \ln\left(\frac{n_i}{n_{tot}}\right) + \sum_{j=1}^{k} \lambda_j a_{ij} = 0 \tag{7}$$

where L is the Lagrange function and λ_j is the Lagrange multiplier. The above equation represents n equilibrium equations of each species present in the system. Consequently, for each species present in the system an equation is taken from Eq. (7) the remaining equations for Langrange multiplier comes from the mass constraint equations.

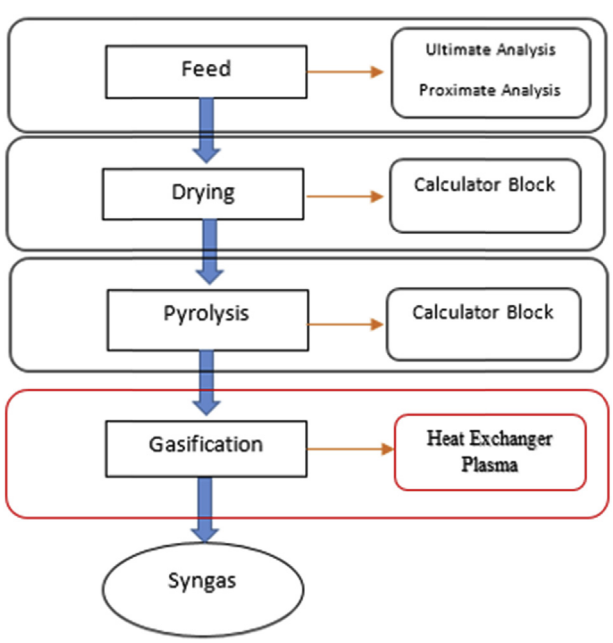

Fig. 2 – Plasma gasification process flow sheet.

Process modelling with Aspen Plus

The simulation operation flow begins with the fuel inlet through the current FEED1 which enters the drier. Here the humidity of the fuel is removed by making use of a calculator block WATER CALCULATOR and a Fortran subroutine. The fuel and moisture flow through FEEDH2O leading to the first separator, SEP1, where moisture is removed from the process. Dried fuel is routed to the DECMPOSE in current DRYFEED. In the DECMPOSE block, using the calculator block YIEL CALCULATOR, and a Fortran subroutine, the fuel is converted into its constituents C, O, H and N, which is introduced in the gasifier by the flow TO-GASIF. At the same time, air (AIR) and steam (STEAM) flows are introduced in the heat exchanger (TORCH), where the plasma arc is simulated. The generated plasma flow (PLASMA), enters the gasifier as a gasification agent. In the gasifier (GASIFY), the gasification process occurs and the software calculates the gas composition which is carried out of the gasifier by the GASIFEID flow into the last separator (SEP2), where the ashes are removed cleaning the syngas (GAS) flow. The diagram of

operation of the simulated process in Aspen Plus can be seen in Fig. 3.

The fuel inlet in the model is carried out by the flow FEED1. The fuel is defined in Aspen Plus as an unconventional solid and its characteristics calculated by the software using the proximate and ultimate analysis of the fuel provider by the user.

The DRIER, a stoichiometric block in Aspen Plus, is responsible for moisture removal. The fuel in the FEED1 flow enters the DRIER block, which together with the WATER CALCULATOR block perform the separation into the fuel and moisture streams, and then introduced into the separator (SEP1) trough the stream FEEDH2O.

In order to calculate the unconventional solid fuel drying, Aspen Plus requires a calculation block able to identify the moisture of the fuel given by proximate analysis. A FORTRAN subroutine corresponding to the drying block was taken from Aspen Plus manual [21].

The DECMPOSE block simulates the pyrolysis. This block, a RYield in Aspen Plus, transform the unconventional solid fuel (biomass) into its elements (C, H, O, N, S, etc.). The molar fraction of the biomass elements are calculated making use of proximate and ultimate analysis and a calculator block, YIELD CALCULATOR, previously defined by the user in FEED1.

It is assumed that the total yield of volatiles is equal to the volatile content of the biomass. The yield distribution of biomass into its components was specified by a FORTRAN subroutine in calculator block.

The GASIFY block is responsible for simulating the gasification process. This reactor is used to model the gasification reactions since it handles with three phase chemical equilibrium. The reactor calculates the syngas composition by minimizing the Gibbs free energy. The decomposed biomass and air enter into the RGibbs reactor where the gasification reactions occur. All the sulfur in the biomass reacts with hydrogen to form hydrogen sulfide. The inaccuracy of this assumption is negligible due to the low contents of sulfur in the biomasses. It was also assumed that only ammonia forms and no nitrogen oxides are produced; an assumption that has already been used in other works [22,23].

The SEP1 and SEP2 blocks are responsible for the removal of moisture and ash, respectively. This type of blocks with the designation Sep in Aspen Plus, perform the separation of a stream based on specified split fractions defined by the user. In the case of SEP1, the stream coming from the drier block FEEDH2O, containing fuel and moisture, is separated into moisture and fuel being the former completely removed from the process. Ash separation from the syngas has been simulated by unit operation model Sep (SEP2). The outlet stream of the RGibbs reactor enters into the SEP2 block which separates gases from ash based on specified split fractions.

The plasma arc is simulated in the model by the TORCH block, which is a heater block in Aspen Plus. This block acts as a heat exchanger where the power of the plasma torch is user defined.

Model assumptions

The model is based on the following main assumptions:

- Stationary state and zero-dimensional;
- The process is considered isothermal. Only the gasifier temperature will influence the gasification reactions;
- The HCOALGEN and DCOALIGT property models were chosen to estimate the biomass enthalpy of formation, specific heat capacity and density based on the ultimate and proximate analyses;
- The chemical species used in the model are: H_2, O_2, N_2, CO, CO_2, CH_4, H_2O and C;
- The plasma torch has a power of 40 kW;
- Ash is a non-reactive nonconventional solid.

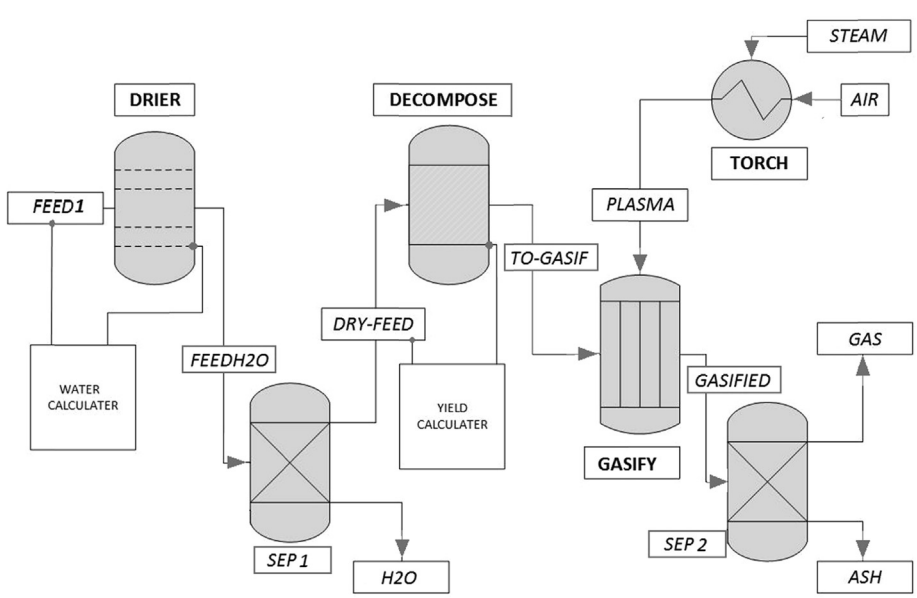

Fig. 3 – Aspen Plus plasma gasification scheme.

Model validation

The developed model was validated against published data by Diaz et al. [24] for plasma gasification. The model inputs were the same as those presented in Table 1.

Fig. 4 shows the values obtained by the model developed in comparison with the literature values.

As we can see from Fig. 4, the values obtained by the developed model shows good agreement with those obtained experimentally by Diaz et al. [24].

Application

The validated model was employed to perform sensitivity analyses of the main operating variables with respect to gasifier performance. Important parameters including temperature, ER and SBR were varied over a wide range. The sensitivity analyses were performed for three biomasses: forest residues, vines pruning and coffee husks and with a single parameter being varied at any given time. The characteristics of the biomasses used in the sensitivity analysis, proximate and ultimate analysis, can be found in Table 2.

Effect of equivalent ratio

According to Niu et al. [26], ER is defined as the ratio of the oxygen to fuel divided by the oxygen under stoichiometric conditions to fuel with complete combustion of fuel:

$$ER = \frac{oxygen/fuel\ mass}{stoichiometric\ oxygen/fuel\ mass} \tag{8}$$

ER was varied between 0.1 and 0.6, while setting a gasifier temperature of 1000 °C and a biomass flow rate of 8 kg/h. Figs. 5–7, show the evolution of the molar fraction of syngas by increasing the amount of air admitted to the system.

It can be observed that when ER rose, the CO_2 content increased, while CO was reduced, as oxidation reactions were favored. A decrease in H_2 was also observed for the three biomasses, because the oxidation reactions were favored when the reaction medium had higher contents of oxygen. The increase in ER also means that more air is introduced in the process. Therefore, it is not surprisingly that the N_2 content increases. This is explained by the fact of N_2 being an inert gas for the temperature level selected. The high N_2 content is a disadvantage since it reduces the syngas LHV. Similar results can be found in the literature [27,28].

The decrease of CO and H_2 concentrations is influenced by the temperature values, since a higher temperature is favorable for endothermic reactions, promoting reduction reactions that lead to the decrease of these elements as indicated by the water-gas shift reaction (Reaction (9)) [29].

The initial oxidation reaction promotes the disappearance of the O_2 which is transformed into CO_2, and some of this element is used in the Boudouard reaction (Reaction (10)) [29] where it is consumed and subsequently produced novel compounds CO_2, justification this assigned the increase of CO_2 over the increase in ER, and the increased N_2 is explained by the influence that this compound has the dilution of other elements [30,31].

$$C + H_2O \leftrightarrow CO + H_2 + 131\ kj/mol \quad \text{Water} - \text{gas shift reaction} \tag{9}$$

$$C + CO_2 \leftrightarrow 2CO + 172\ \text{kj/mol} \quad \text{Boundouard reaction} \tag{10}$$

Effect of steam-to-biomass ratio

The use of steam as gasifying agent is used in the enrichment of the mole fraction of H_2 in the final product [32]. The steam to biomass ratio (SBR) is defined as the steam mass flow rate divided by the biomass mass flow rate in dry basis:

$$SBR = \frac{Steam\ mass\ flow}{Biomass\ mass\ flow} \tag{11}$$

The SBR was varied over a range of values from 0.5 to 2.0, while the gasifier temperature set at 1000 °C and a biomass flow of 8 kg/h. Figs. 8–10 shows, respectively, the syngas molar fractions as a function of the SBR for the three biomasses under study.

As expected, the introduction of steam in the process results in an increase of H_2 molar fraction in syngas. The use of steam as a gasifying agent increases the partial pressure of water vapor inside the gasification reactor which favors the water gas shift and steam reforming reactions, leading to increased H_2 production [33]. The presence of steam in gas-phase reactions results in the decomposition of hydrocarbons and increasing contents of H_2 and CO_2 as reaction products, which is confirmed for the three biomasses under study. The results obtained agree with those found in literature [34].

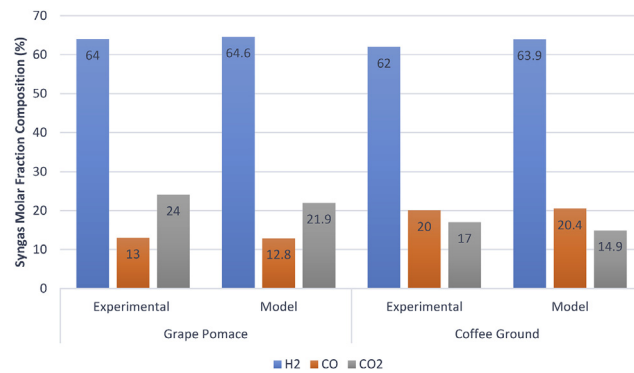

Fig. 4 – Comparison of Aspen Plus and experimental results.

Table 1 – Operating parameters for model validation [22].

	Grape pomace	Coffee ground
Biomass (kg/h)	4.2	4.2
Steam (kg/h)	12	28
Air (kg/h)	0	0
Gasifier temperature (°C)	1720	1220

Table 2 – Biomass properties [25].

Biomass properties	Forest residues	Coffee husk	Vines pruning
Ultimate analysis:			
N	2.4	5.2	2.6
C	43.0	40.1	41.3
H	5.0	5.6	5.5
O	49.6	49.1	50.6
Humidity	11.3	25.3	13.3
Density (kg/m^3)	650	500	265
Net heat value (MJ/kg)	21.2	20.9	15.1
Proximate analysis:			
Ash	0.2	2.5	3.1
Volatile matter	79.8	83.2	83.6
Fixed carbon	20.0	14.3	13.3
Chemical formula	$C\,H_{1.510}O_{0.61}N_{0.002}$	$C\,H_{1.676}O_{0.918}N_{0.111}$	$C\,H_{1.598}O_{0.919}N_{0.054}$

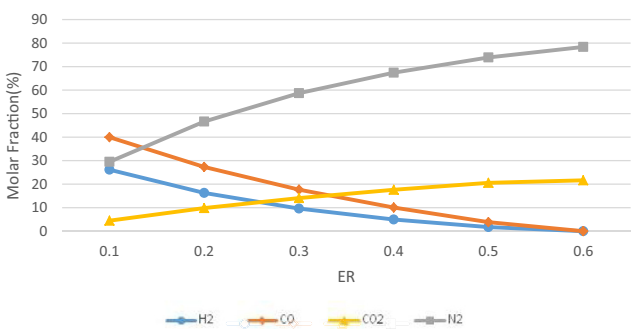

Fig. 5 – Effect of the ER on syngas composition for forest residues.

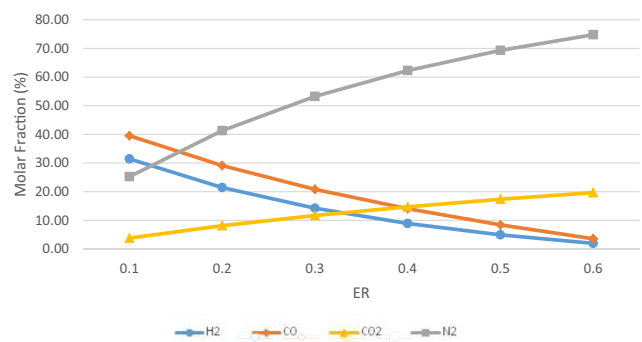

Fig. 7 – Effect of the ER on syngas composition for coffee husks.

Effect of gasifier temperature

One of the most important parameters in the gasification process is the gasifier temperature. The temperature of the gasifier is the major participant in the composition of the synthesis gas, since there are certain gasification equations that require specific temperatures. Setting an ER of 0.3 and a SBR of 0.5, the temperature inside the gasifier was varied between 900 °C and 2000 °C. Figs. 11–13 show the syngas molar fraction as a function of the gasification temperature for the three biomasses under study.

It can be seen from Figs. 11–13 that the increase of the gasification temperature promotes the formation of a syngas with higher CO content. On the other hand, the H_2 and CO_2 contents follow an opposite trend. This variation of the concentration values is due to the nature of the water shift reaction being exothermic. The high temperature of the gasifier does not favor the reaction leading to lower molar fractions of products, CO_2 and H_2, the opposite response to the increase in steam introduced into the system as in the case of SBR study. The CO as a water shift reaction reagent undergoes the inverse result, with the increase in temperature favoring its molar

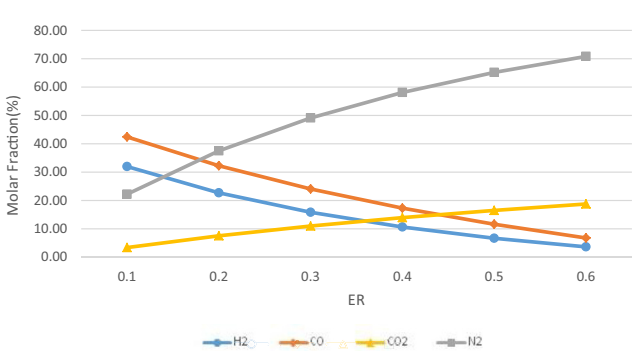

Fig. 6 – Effect of the ER on syngas composition for vines pruning.

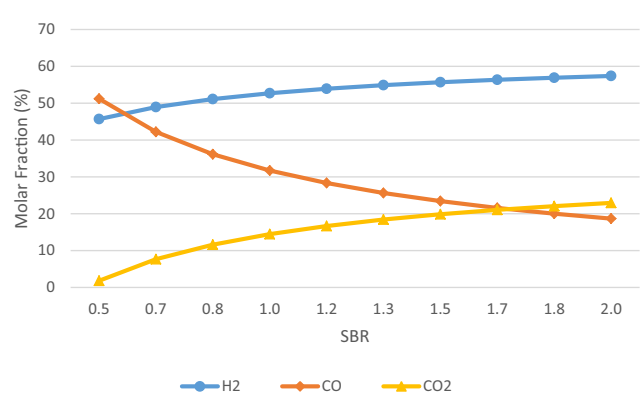

Fig. 8 – Effect of the SBR on the syngas composition for forest residues.

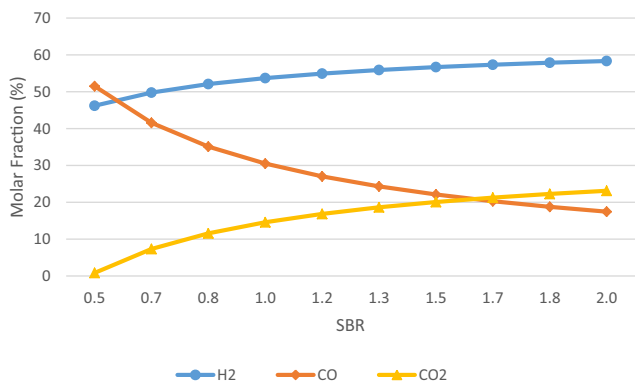

Fig. 9 – Effect of the SBR on the syngas composition for vines pruning.

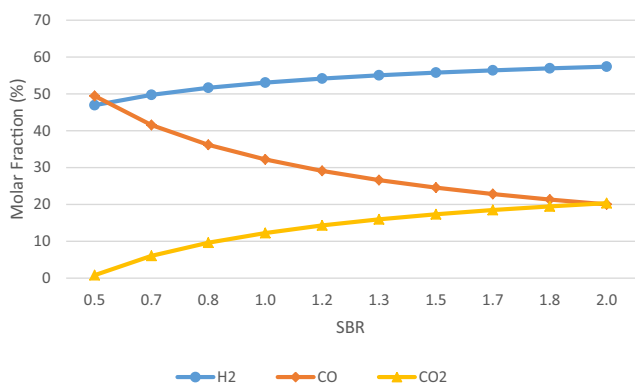

Fig. 10 – Effect of the SBR on the syngas composition for coffee husks.

fraction. These results are in agreement with the literature [28].

Comparing the 3 figures it is verified that there is similarity to the behavior of the molar fraction on temperature effect, that is, the molar fraction of CO and N_2 increases and the fraction of CO_2 and H_2 decreases along the 3 graphs. However, there are minimal differences regarding the specific values for each temperature, since biomass represented in each figure has different molar compositions, so its molar fraction will be different in each graph.

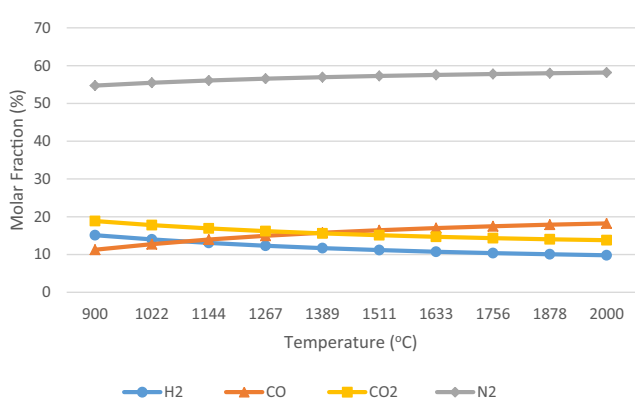

Fig. 11 – Effect of the temperature on the syngas composition for forest residues.

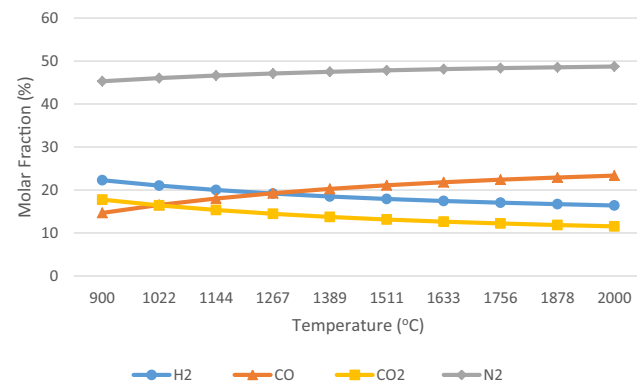

Fig. 12 – Effect of the temperature on the syngas composition for vines pruning.

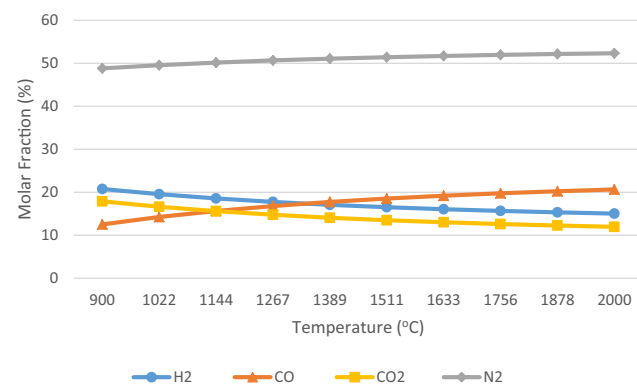

Fig. 13 – Effect of the temperature on the syngas composition for coffee husks.

Hydrogen production

Combining two of the parameters studied previously, SBR and temperature; the variation of the molar fraction of H_2 in syngas was studied. The temperature was varied between 900 °C and 2000 °C and the SBR between 0.2 and 2.0.

Figs. 14–16 show the results obtained for the three biomasses under study. The molar fraction of H_2 in the final product is always greater than 45% for all the cases. However, coffee husks have a higher prospect for H_2 production because

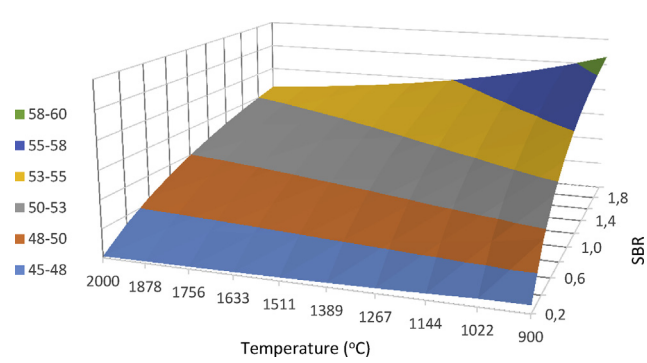

Fig. 14 – Effect of the SBR and temperature on the H_2 molar fraction of forest residues gasification.

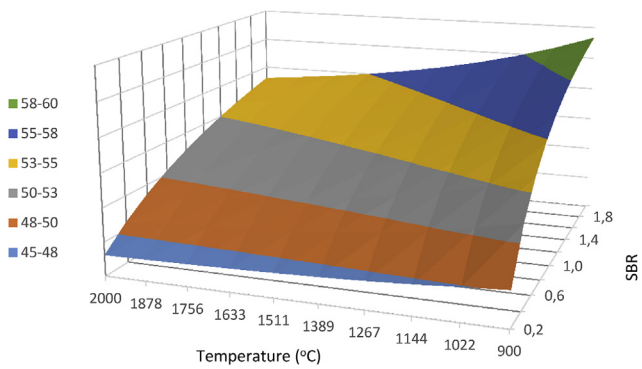

Fig. 15 – Effect of the SBR and temperature on the H_2 molar fraction of vines pruning gasification.

of the higher range of parameters that allow the maximum H_2 production.

Lower heating value

The lower heating value (LHV) is defined as the amount of heat released by fully combusting a fuel less the heat of vaporization of the water in the combustion product. Kaewluan and Pipatmanomai [35] propose the following equation to calculate LHV of a gas:

$$LHV_{Gas} = 10.79Y_{H_2} + 12.62Y_{CO} + 35.81Y_{CH_4} \quad (12)$$

where Y represents the volume fraction of each component in the syngas on a dry basis.

The following Figs. 17–19 represent the variation of the syngas LHV with the parameters studied previously, ER, SBR, and gasifier temperature for the three biomasses under study.

From these figures it is possible to conclude that both ER and SBR have negative effect on syngas LHV, while the temperature has a positive effect on syngas LHV. The explanation for this behavior can be based on the H_2 and CO contents variation in the syngas. Both H_2 and CO molar fractions decrease with ER as seen in Figs. 5–7. CO molar fractions decrease with SBR, while H_2 molar fraction increases (Figs. 8–10). CO molar fraction increases with temperature, while H_2 molar fraction decreases (Figs. 11–13). As expressed by Eq. (12) there is an higher influence of CO than H_2 in the LHV.

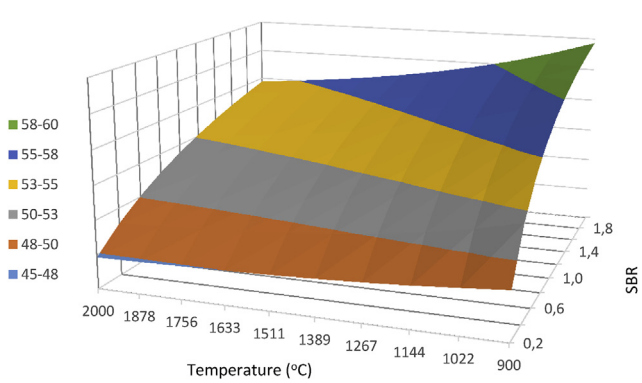

Fig. 16 – Effect of the SBR and temperature on the H_2 molar fraction of coffee husks gasification.

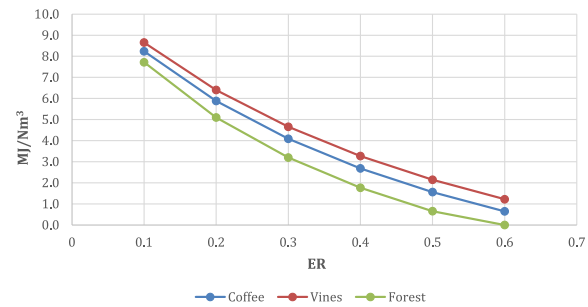

Fig. 17 – Effect of the ER on the syngas LHV.

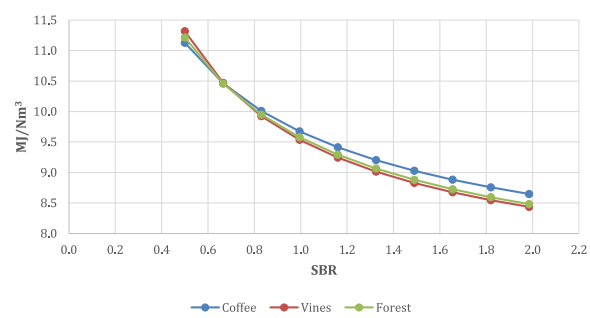

Fig. 18 – Effect of the SBR on the syngas LHV.

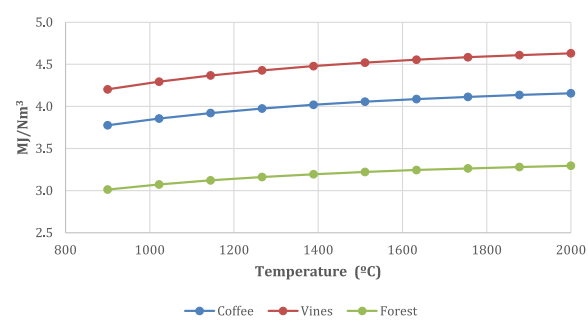

Fig. 19 – Effect of the temperature on the syngas LHV.

Therefore, syngas LHV should decrease with increasing ER and SBR and increase with increasing temperature.

Conclusion

A process model was developed to simulate the biomass plasma gasification using Aspen Plus simulator. The model was used to optimize plasma gasification with air and steam of three biomasses. The results are presented in term of product gas composition along with the LHV of the product gas.

The results indicate that low ER is more favorable for the production of hydrogen for the three biomasses, because the oxidation reactions are favored when the reaction medium had higher contents of oxygen.

The results show that the low temperature is more favorable for production of hydrogen. The simulation results also

demonstrate that steam injection is a key factor to produce more hydrogen rich gas in the SBR range studied, but had a major effect on CO_2 formation. The temperature and the SBR show opposite behavior on the syngas LHV, which is attributed to the CO content in the syngas that increases with temperature and decreases with SBR.

Results of plasma gasification show similar syngas LHV trends for the three biomasses cases. Higher syngas LHV is obtained with vines pruning. These data are crucial to describe scenarios concerning the potential use of biomass as a source of energy.

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